# M. Waseem Ashraf

Email : waseemashraf1584@gmail.com
https://www.linkedin.com/in/m-waseem-ashraf/
Portfolio : https://mwaseemashraf.github.io/

Professional Summary

PhD Mechanical Engineer with 4+ years of experience in hardware and electromechanical system design, specializing in thermo-mechanical and multiphysics FEA. Excited to apply my knowledge in ASIC design.

Technical Skills

- **Software**: ANSYS Mechanical, ANSYS Maxwell (HFSS), SolidWorks, COMSOL Multi-physics, Microsoft Office.
- **Programming Languages**: Python, MATLAB, C++, Batch Scripting, SLURM (HPC).

Experience Highlights

- **Brigham Young University** — Provo, UT
  Graduate Research Assistant — *Aug 2021 - Present*
  - **Thermo-Mechanical Design & FEA Optimization**: Supported the design and validation of a zirconia-based solid-state actuator; partnered with physicists and materials scientists to evaluate FEA-informed mechanical and thermal trade-offs under manufacturing constraints, resulting in a 40% energy-efficiency gain and a 105 K lower operating temperature.
- **Lahore University of Management and Sciences** — Lahore, Pakistan
  Hardware Design Engineer — *Aug 2018 - Aug 2021*
  - **High-Speed Electromechanical System Design & FEA Validation**: Performed thermo-mechanical FEA of a compact, air-cooled electromechanical assembly with high-RPM motors and embedded electronics to evaluate thermal performance, stress, and structural stability; partnered cross-functionally to drive DFT/DFA design iterations and deliver a solution meeting all customer and manufacturing requirements.
  - **Advanced FEA of High–Aspect-Ratio Multilayer Structures**: Led the finite-element simulation of a first-of-its-kind four-layer (skin/fat/bone/brain) human head model with extreme thickness ratios, resolving numerical accuracy and mesh-stability challenges analogous to MCM and flip-chip BGA package geometries; collaborated with physicists to calibrate electromagnetic–thermal FEA against first-principles formulations and power-balance checks, enabling the first published four-layer realistic head model for SAR and temperature prediction.
  - **Hyperloop Pod FEA Passive Levitation Optimization**: Served as an FEA lead and later team head for university Hyperloop pod teams (SpaceX Hyperloop Pod Competition, an annual global student design/build event) where I used ANSYS Maxwell to perform finite-element simulations of passive magnetic levitation systems, optimizing magnet orientation, geometry, and pod speed parameters; validated FEA results with analytical Maxwell equations and cross-functional design integration to improve levitation performance under strict design constraints.
- **Freelance Mechanical Design Work** — Remote
  Contracted Design Engineer — *Jan. 2017 - Jun. 2021*
  - **Portfolio Overview**: Executed hardware design and validation using FEA (Finite Element Analysis) for individual and corporate clients. Completed 20 projects over 4.5 years, with project timelines ranging from three weeks to four months and an average client rating of 4.9/5.

Education

- **Brigham Young University, BYU** — Provo, UT
  PhD in Mechanical Engineering; GPA: 3.75/4.00 — *Sep. 2021 – May. 2026(Expected)*
- **Pakistan Institute of Engineering and Applied Sciences, PIEAS** — Nilore, Pakistan
  BS in Mechanical Engineering; GPA: 3.91/4.00 — *Sep. 2014 – Jun. 2018*

Leadership Experience

- **BYU**: President, Official Mechanical Engineering Graduate Association (OMEGA), 2024–2025.
- **BYU**: Vice-President, OMEGA, 2023–2024.
- **BYU**: Graduate Delegate, College of Engineering – Graduate Student Society, 2021–2022.

Achievements & Awards

- **The Minerals, Metals & Materials Society (TMS) Conference**: Research presentation award–2025.
- **BYU**: Recipient of Ira. A. Fulton Fellowship–2021.
- **Rhodes Scholarship, Oxford University**: Finalist for the Rhodes Scholarships Oxford University–2018.